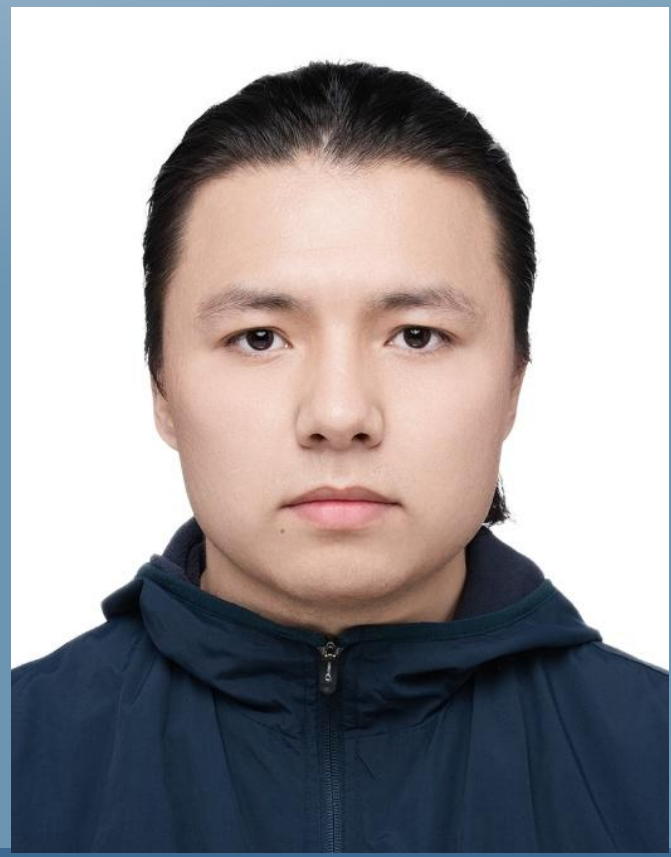




EVOLUTION OF STAR CLUSTERS WITH INITIAL BULK ROTATION VIA N-BODY SIMULATIONS

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ABSTRACT

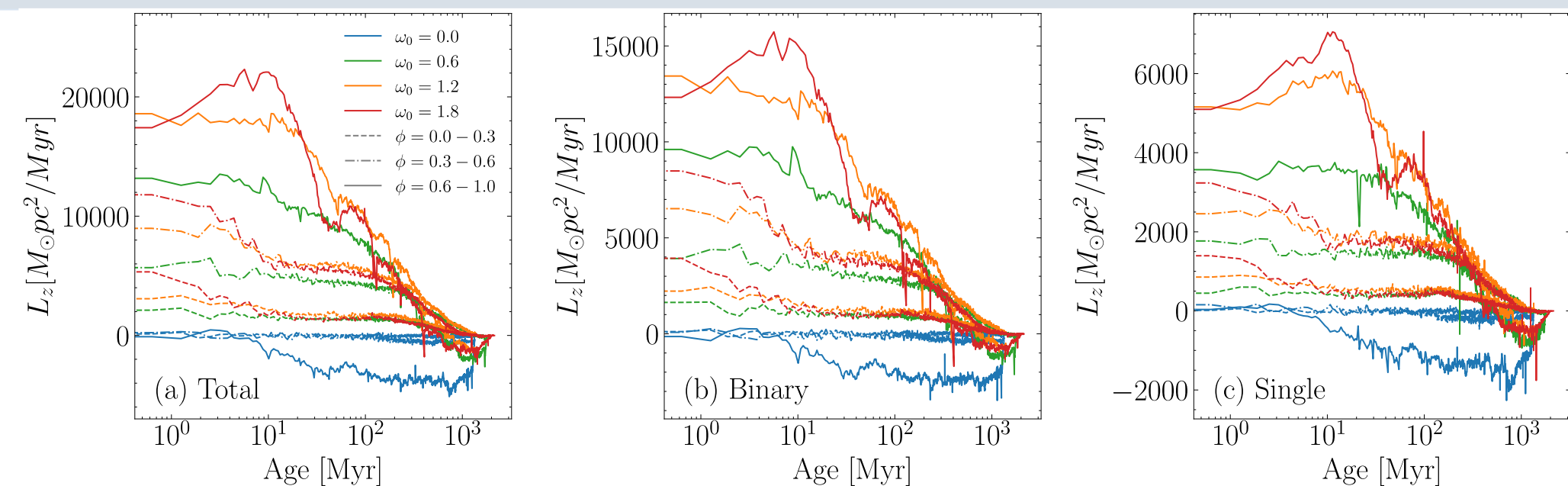
Young star clusters can inherit bulk rotation from the molecular clouds from which they have formed. This rotation can affect the long-term evolution of a star cluster and its constituent stellar populations. In this study, we aim to characterize the effects of different degrees of initial rotation on star clusters with primordial binaries. The simulations were performed using NBODY6++GPU. Rapidly rotating clusters show angular momentum transport from the inner parts to the outskirts, resulting in a core collapse. Angular momentum transport is accompanied by a highly elongated bar-like structure morphology. Rotating and nonrotating clustersexperience changes in the direction of angular momentum near the dissolution and early evolution due to the tidal field, respectively

INTRODUCTION

The dynamics of stellar systems is a major research field of astrophysics (Binney & Tremaine, 2011; Chandrasekhar, 1960; Krumholz et al., 2019). The study of stellar dynamics can provide constraints on the conditions of star formation and can be used to examine the effects of properties on the system. One of such properties is bulk rotation, which is the rotational motion of a star cluster around its center of mass. Star clusters inherit initial rotation from molecular clouds through the star formation process (Lahén et al., 2020; Mapelli, 2017). At later times, bulk rotation can evolve through tidal interactions with other star clusters and tidal interactions with the Galactic tides. Over time, bulk rotation can significantly influence the morphology, core collapse process, and long-term evolution of star clusters.

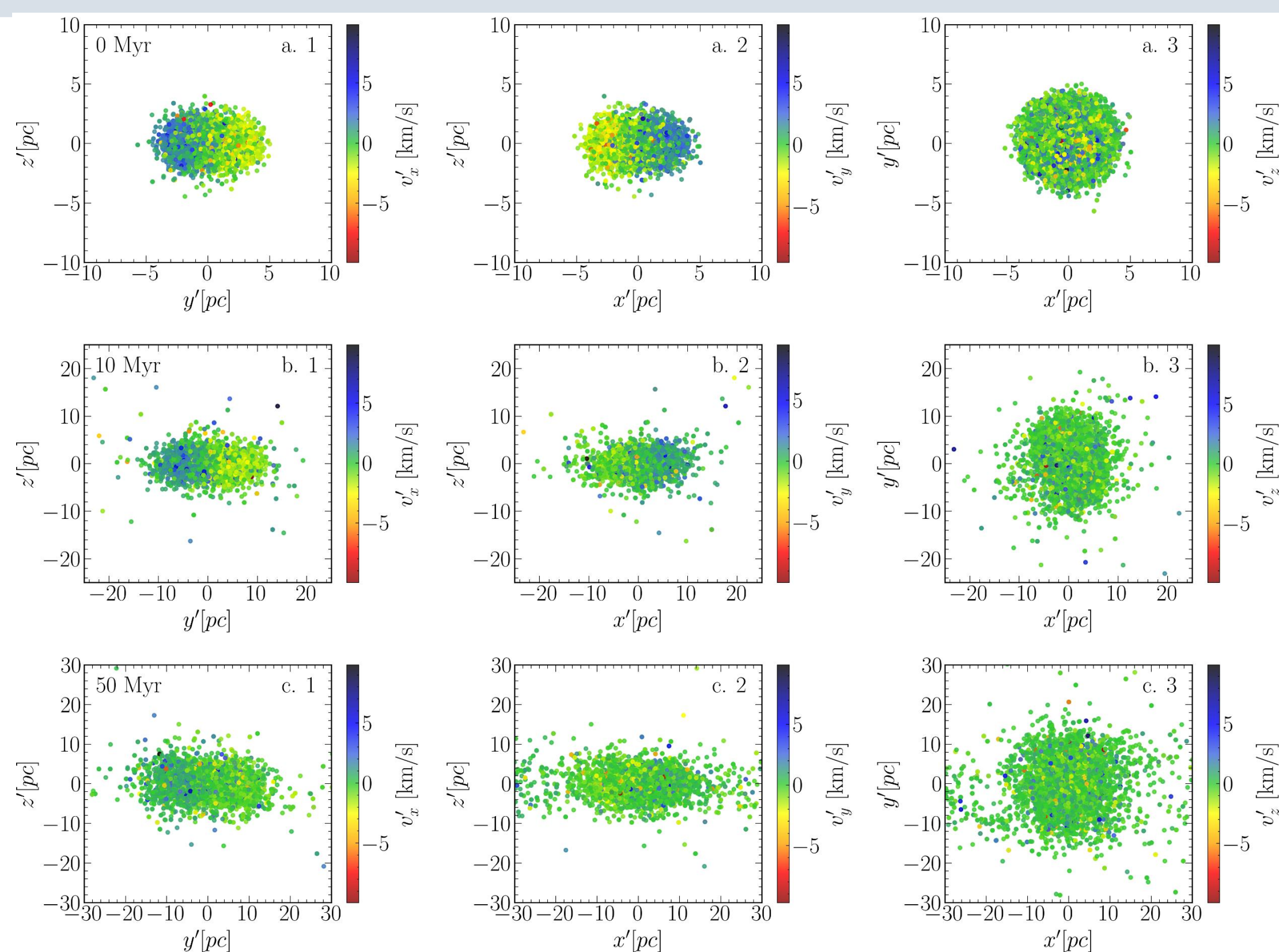
RESULTS

We quantify the rotation signature by examining the angular momentum in the **inmost (dashed)** and **intermediate (dot dashed)** side to the **outmost (solid)** parts of the system.



- We could notice **the angular momentum transport** from the inmost and intermediate parts to the outmost.
- This is the sign of **gravothermal-gravogyro catastrophe**.
- There is eventual change of angular momentum **from the positive sign to negative** in all the models.

We transformed the coordinates into the angular momentum vector's direction at each time and inspect stars by their velocities such that the net angular momentum of the star cluster points in the direction of the z' -axis.



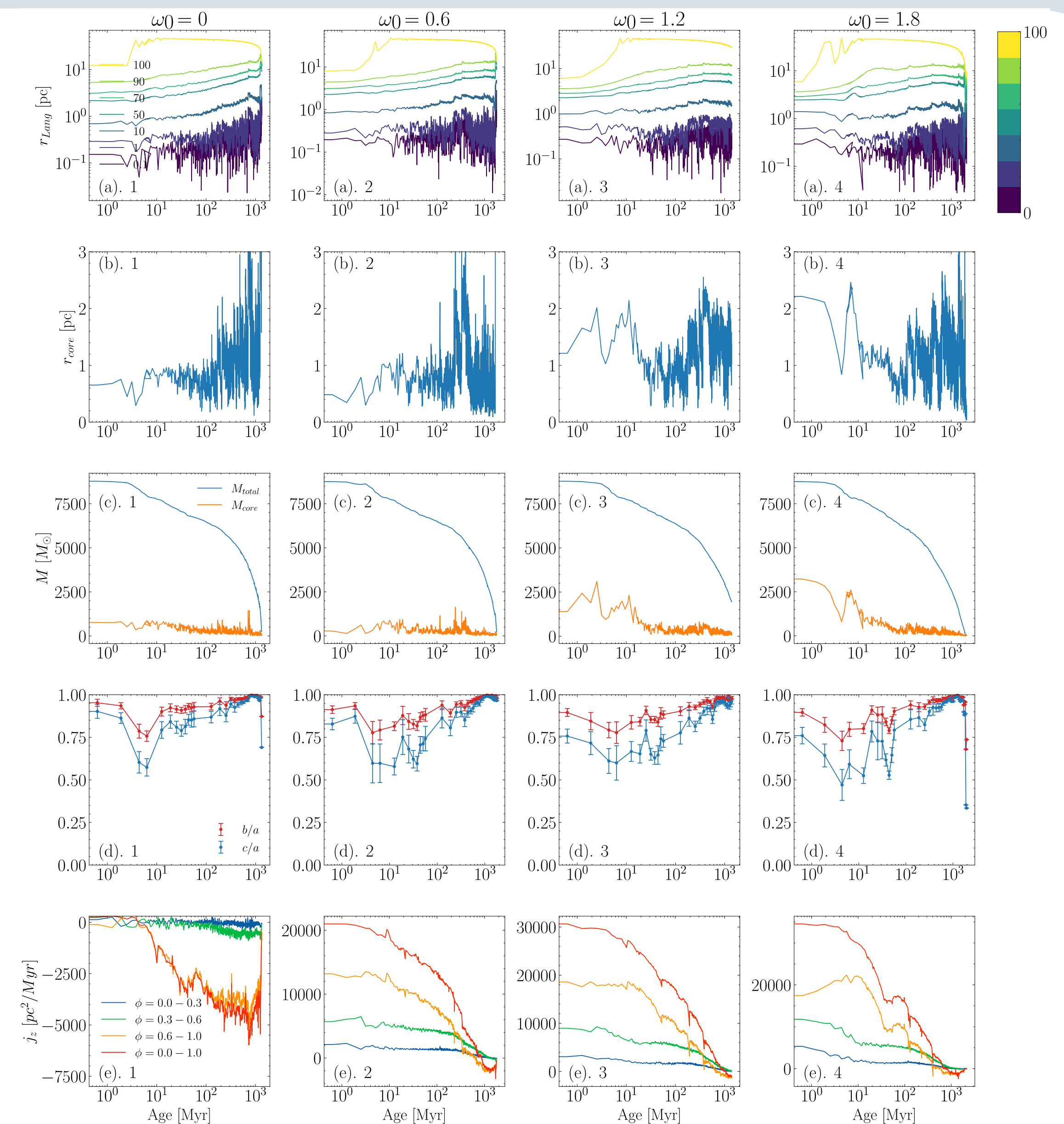
CONCLUSION

- There is angular momentum transport from the inner part of to the outer in the models with high bulk rotation. There is also the core-collapse at the time of angular momentum transport. This indicates the Gravothermal and Gravogyro catastrophe that lead to core-collapse or heavy core contraction of the clusters.
- There is a change in angle of angular momentum to completely opposite direction seen early in models with no bulk rotation and later in models with bulk rotation.
- Bulk rotation affects the long-term dynamic evolution and strongly affects the morphology at the early stages of evolution. The effect is seen in high elongation at the timescales of angular momentum transport.

MODELS

We simulate **20 star cluster simulations** using NBODY6++GPU in order to study the effects of bulk rotation:

Models	N	f_{bin}	Z	ω_0	r_h	r_t
N10k_fb00_z001_w000	10 000	0%	0.01	0.0	2.25	21.65
N10k_fb00_z001_w006				0.6	2.25	21.65
N10k_fb00_z001_w012				1.2	2.23	21.65
N10k_fb00_z001_w018				1.8	2.45	21.65
N11k_fb01_z001_w000	11 000	10%	0.01	0.0	2.03	22.38
N11k_fb01_z001_w006				0.6	2.18	22.38
N11k_fb01_z001_w012				1.2	2.23	22.38
N11k_fb01_z001_w018				1.8	2.40	22.38
N11k_fb01_z002_w000	11 000	10%	0.02	0.0	2.03	22.38
N11k_fb01_z002_w006				0.6	2.18	22.38
N11k_fb01_z002_w012				1.2	2.23	22.38
N11k_fb01_z002_w018				1.8	2.40	22.38
N15k_fb05_z001_w000	15 000	50%	0.01	0.0	2.09	24.82
N15k_fb05_z001_w006				0.6	2.25	24.82
N15k_fb05_z001_w012				1.2	2.36	24.82
N15k_fb05_z001_w018				1.8	2.41	24.82
N15k_fb05_z002_w000	15 000	50%	0.02	0.0	2.09	24.82
N15k_fb05_z002_w006				0.6	2.27	24.82
N15k_fb05_z002_w012				1.2	2.36	24.82
N15k_fb05_z002_w018				1.8	2.41	24.82



We examine the global parameters such as Lagrangian radii r_{lagr} , core radius r_{core} , total mass M_{total} and core mass M_{core} , principal axes ratios b/a and c/a , and specific angular momentum along the z -axis of the model with 15k stars, $f_{bin}=50\%$, $Z=0.01$, for different values of ω_0 .

- The core collapse is seen in the drastic changes in r_{core} and M_{core} among all the models. The models with high rotation experience it more drastically and earlier than the models with no or moderate rotation.
- At the time of said core collapses, there is a **significant elongation** and **bar like structure** that can be noticed from principal axis ratios.
- Said time also coincides with the angular momentum transport from the inmost and intermediate regions to the outmost.

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